

Multitask Learning for Tree Instance Segmentation and Diameter Estimation from RGB-D Imagery in Agroforestry and Rainforest Environments

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Abstract

Tree inventory is a pivotal process in forest management, and it relies on tree identification and parameters such as diameter at breast height (DBH) and height. Small errors in measurement or identification can lead to miscalculation of tree volume and carbon accounting. In Indonesia, inventory is mostly carried out manually due to the high cost of terrestrial laser scanning and limited imagery of local tree species. The purpose of this research is to train MaskDINO, a computer vision model, on four-channel images, leveraging the architecture of the vision foundation model FocalNet-L as its backbone. The model was trained using SPREAD, a synthetic dataset. The segmentation head identifies tree species, while a second head estimates DBH from backbone features at the image row corresponding to breast height. Hence, a single model can identify tree species and estimate DBH from ground-level RGB-D images captured under the canopy. This research shows that the final model, trained with diameter supervision across eleven species, achieves a test AP₅₀ (a standard measure of object-detection accuracy) of 62.29 %, with no measurable cost to segmentation compared with an otherwise identical segmentation-only model. In an end-to-end evaluation using the model’s own detections and predicted masks rather than ground-truth annotations, the model successfully estimates diameter for 27.1 % of eligible trees, with a DBH root-mean-square error of 7.62 cm. In a controlled comparison where both models are trained from scratch and differ only in channel count, adding depth increases AP₅₀ by 12.92 percentage points for FocalNet-L, demonstrating the importance of depth information for tree detection. Detection recall falls with camera distance, from 82.4 % for trees within 5 m to 34.5 % for trees beyond 30 m. This indicates that the model detects nearby trees reliably but increasingly misses distant and occluded trees, while species classification and relative diameter error remain largely stable for detected trees.

Keywords: forest inventory, instance segmentation, RGB-D, multitask learning, diameter at breast height, synthetic data

1. Introduction

Tropical and plantation forests in Indonesia are among the most biodiverse and carbon-dense ecosystems in the world, playing a critical role in global climate regulation (Baccini et al., 2017). Managing these forests rests on forest inventory,

which turns standing vegetation into numbers that policy can act on. Carbon stock estimates, biodiversity indices, and timber yield models are all built from per-tree records, of which species, trunk diameter, and height matter most, and this study addresses the first two. Diameter at breast height (DBH) enters allometric biomass equations directly (Manuri et al., 2016), so error in DBH propagates into every downstream carbon figure (Baccini et al.,

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2017). Across Indonesian forests and agroforestry systems, these records are still collected by hand with diameter tapes. Its cost lies in access, repetition, and the skilled labour needed to identify species in the field, and it scales with the number of trees rather than the area surveyed.

Two technologies have been proposed to reduce that cost. Terrestrial laser scanning (Liang et al., 2018) recovers trunk geometry with millimetre precision but requires equipment whose price places it outside routine operational inventory in much of the region. Photogrammetric and image-based approaches use commodity cameras, but supervised learning needs annotated imagery, and per-instance annotation of tropical species under closed canopy barely exists. That deep learning can segment trees is well established, at least in aerial and plantation settings. Two harder questions remain. Can it do so from the viewpoint a field crew occupies beneath the canopy? Can it deliver the second measurement that inventory needs, rather than delineation alone?

Prior approaches differ mainly in acquisition protocol. Close-range work captures one trunk from a metre or two, with a smartphone photograph (Wu et al., 2019; Holcomb et al., 2023; Xiang et al., 2025), a coarse depth map (Feng et al., 2024), or monocular SLAM (Su et al., 2024; Fan et al., 2018), and reaches errors of a few centimetres. That accuracy is real, but the operator chooses the viewpoint and isolates the trunk before the model runs. Accumulating many views per trunk, from an under-canopy UAV (García-Pascual et al., 2025) or a 360-degree camera carried through the stand (He et al., 2025), improves accuracy the same way, by spending more effort per tree. Work on whole scenes instead tends to deliver one attribute rather than two, whether detection and species without diameter (Duclos et al., 2026), trunk detection in urban scenes (Li et al., 2023b), diameter in planted stands where trunks are separable (Lu et al., 2026), or aerial crown delineation from a viewpoint where trunks are not visible (Ball et al., 2023). The closest precedent is Grondin et al. (2022), which detects multiple trees and estimates diameter from ground-level synthetic renders without resolving species, the

axis this study adds. What remains open is the joint prediction of species-resolved instance masks and trunk diameter for every tree in a single passive under-canopy frame, in stands where trees are neither isolated nor easily separable.

The under-canopy viewpoint complicates both instance segmentation and diameter estimation. Aerial and unmanned-aerial imagery observes crowns from above, where individual trees are largely separable and occlusion is limited to overlapping crown margins. Airborne segmentation procedures recover more than 90% of overstory trees but barely 60% of understory trees, because the upper canopy occludes what lies beneath (Hamraz et al., 2017). Under-canopy imagery inverts that problem rather than removing it. It observes stems and crowns laterally, so trees occlude one another heavily and apparent size varies by more than an order of magnitude with distance to the camera. The trunk at breast height, the attribute inventory most needs, is frequently hidden behind foreground vegetation. Tropical stems compound this, since large trees are often buttressed and markedly non-circular, which the geometric fitting traditionally used for diameter estimation cannot accommodate (Nölke et al., 2015). Depth sensing addresses this geometry directly, and without a distance to the trunk pixel width cannot be converted to a physical width at all.

This study addresses three questions. The first concerns how much an explicit depth channel contributes to instance segmentation under dense canopy, measured under controls that isolate the channel from the initialisation strategy it is usually confounded with. The second concerns whether a single network can produce species-resolved instance masks and trunk diameter together, and what that coupling costs relative to training the two objectives separately. The third concerns where the errors lie once performance is stratified by distance from the camera.

We answer them with a multitask RGB-D network trained on synthetic under-canopy imagery. Synthetic rendering supplies the per-instance masks, species labels, diameters, and registered depth that the annotation shortfall otherwise

makes unavailable, jointly for the same scenes. We use SPREAD (Feng et al., 2025), which renders ground-level forest scenes in Unreal Engine 5, spanning Indonesian agroforestry species and rainforest vegetation in the same partition. This study leverages the architecture of FocalNet-L (Yang et al., 2022), a vision foundation model, as the backbone of MaskDINO (Li et al., 2023a). Its input convolution is widened to accept depth as a fourth channel, and the model is trained from scratch on this four-channel input for the reason Section 3.2 quantifies. A second head regresses diameter from backbone features sampled at the image row that corresponds to breast height. Segmentation covers all thirteen species in the dataset, while diameter is supervised on the eleven for which ground-truth DBH is recorded. All results are therefore measured on rendered rather than photographed imagery, a limitation stated in Section 4.5. Our contributions are as follows.

1. **A controlled measurement of the depth channel.** Prior comparisons of RGB against RGB-D in this setting are confounded, because four-channel input cannot load three-channel pretrained weights and is therefore usually trained from scratch while its RGB counterpart is pretrained. We report pairs in which both arms are trained from scratch and only the channel count differs.
2. **A multitask architecture that couples segmentation and diameter through shared features.** The diameter head reads backbone features, produced by the same forward pass as the segmentation output, at a geometrically derived breast-height row rather than from pooled instance embeddings. A stop-gradient keeps the coupling one-directional, so the shared representation is shaped by segmentation alone.
3. **Distance-stratified evaluation.** Because depth is an input, every ground-truth instance carries a physical distance that is computed from geometry and is independent of model weights. This lets us bin identical instances across models and separate three abilities that

aggregate metrics conflate, namely finding a tree, naming it, and measuring it.

4. **An explicit account of what does not work.** We report a geometric diameter baseline that fails at this resolution and a loss weighting schedule that degrades segmentation without improving diameter. Each is diagnosed rather than merely reported.

Together, these contributions point toward a low-cost alternative to terrestrial laser scanning. A single commodity RGB-D camera, processed by one network, could let resource-limited forestry agencies and smallholder programs track stand structure and carbon stock without specialised equipment or field taxonomists, once the sim-to-real gap identified in Section 4.5 is closed.

2. Materials and methods

Fig. 1 sets out the study in full before the sections below take each stage apart.

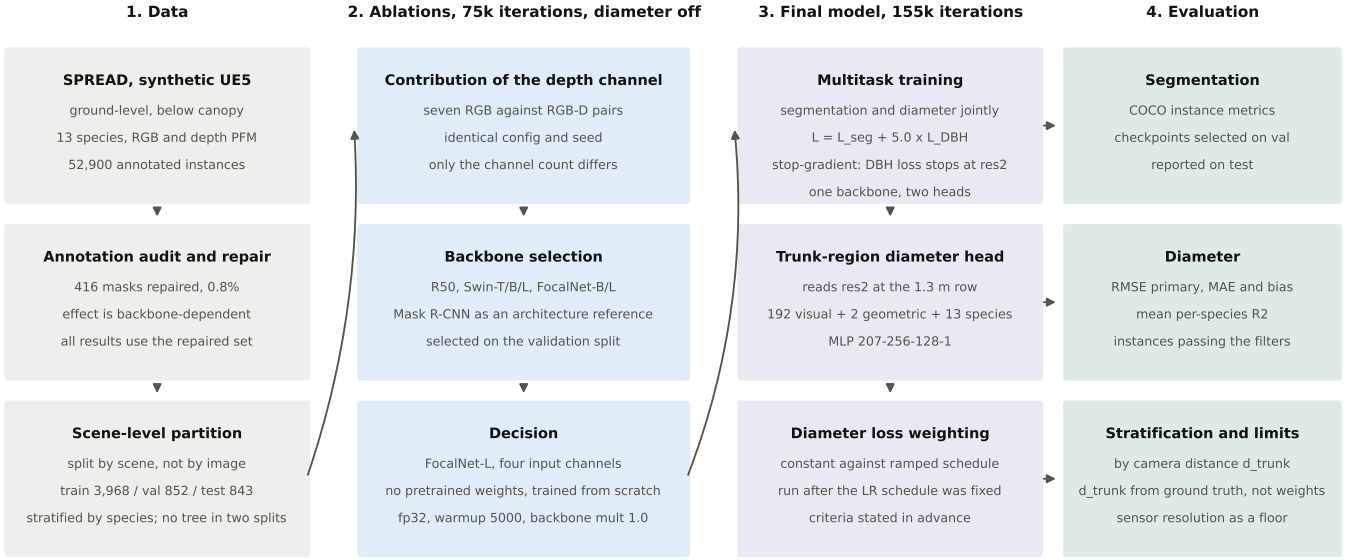
2.1. Material

The subsections below describe the dataset, its partitions, and the species for which diameter is supervised.

2.1.1. Study data

We use SPREAD (Feng et al., 2025), a synthetic dataset rendered in Unreal Engine 5 that simulates ground-level under-canopy views of forest and plantation scenes. Each sample provides a colour image, a per-pixel depth map in metres, per-instance segmentation masks, a species label, and, for a subset of instances, a recorded DBH. The virtual camera stands about 2m above ground and looks laterally into the stand, reproducing the occlusion and distance-dependent scale variation of a field crew’s viewpoint. All imagery is rendered at 480×270 pixels with a vertical focal length of 240 px, a fixed property of the released data that places a floor on attainable diameter accuracy (Section 4.3).

We retain 13 species selected for their occurrence in Indonesian agroforestry and rainforest stands, spanning a plantation domain of orchard



The two training schedules are deliberate and must not be mixed. Ablations run for 75,000 iterations with the diameter objective disabled, so that a full matrix is affordable and channel and backbone comparisons are made on segmentation alone. The final model runs for 155,000 iterations with both objectives active. Values within one ablation table are comparable to each other, but not directly to the final model; the schedule offset is quantified separately.

Figure 1: Study workflow, from data to reported claim. Column 1: the dataset, its annotation audit, and the scene-level partition. Column 2: the two ablations, input representation and backbone, both segmentation-only so a full matrix is affordable. Column 3: the final multitask model and its diameter head. Column 4: evaluation, including the distance stratification.

and fruit species and a rainforest domain of figs, palms, understorey vines, and ferns. This scoping decision aligns the label space with the stands an Indonesian inventory would encounter, at the cost of a narrower taxonomy than the renderer can produce. Table 1 reports the split sizes. The partition is made at the level of the rendered scene rather than the individual image, because one scene yields several images and an image-level split would place the same physical tree on both sides of it, and within that constraint it is stratified by species. An annotation audit preceded all experiments; repairing 416 masks, 0.8% of instances, improved a pretrained RGB configuration by 1.7 points of AP_{50} while leaving an RGB-D configuration trained from scratch unchanged within noise. All results use the repaired annotations.

2.1.2. Species subset for diameter

Diameter supervision covers eleven of the thirteen species. The two excluded carry no recorded DBH for more than half of their instances, leaving nothing to supervise. Segmentation continues to

Table 1: Dataset partitions. Instance counts are annotated tree instances; not all carry a recorded DBH.

Partition	Images	Instances
Train	3968	37 105
Validation	852	7870
Test	843	7925

use all thirteen classes, so the restriction changes only which instances contribute to $\mathcal{L}_{\text{DBH}}$. Supervising every species the annotation supports is a deliberate choice about coverage, since an inventory is more useful the more of the stand it can report a diameter for. The eleven species admit 1790 test instances with a well-posed breast-height geometry inside 20 m. Section 4.5 records what the choice costs, since four of the eleven are small fruit trees whose form makes a single diameter at 1.3 m an imperfect description.

2.2. Method

The subsections below describe where this implementation departs from the reference MaskDINO and FocalNet-L architectures, and introduce two components with no counterpart in either, a trunk-region diameter head (Section 2.2.3) and the training scheme that couples it to the backbone (Section 2.2.4).

2.2.1. Depth as a fourth input channel

Depth enters the network as a fourth channel alongside RGB rather than through a separate encoder. Raw depth is heavily right-skewed, since most stems fall within 10 m while the rendered range extends past 200 m, so a linear encoding would compress the near field into a narrow band of values. We therefore apply a logarithmic transform,

$$d_{\text{norm}} = \frac{\log(1 + d)}{\log(1 + d_{\text{max}})}, \quad d_{\text{max}} = 237.6 \text{ m}, \quad (1)$$

and rescale to the 0 to 255 range used by the other channels. Pixels flagged as non-returns, which correspond to sky, are mapped to zero. The channel statistics used for normalisation are extended accordingly, with the fourth-channel mean and standard deviation set to 118.1 and 50.8.

For segmentation this encoding supplies a cue that separates a near trunk from a similarly coloured background trunk. Its second role, converting an apparent trunk width in pixels into a physical width, is used explicitly in Section 2.2.3.

2.2.2. Segmentation architecture and ablation protocol

We build on MaskDINO (Li et al., 2023a), a query-based instance segmentation architecture that predicts masks, boxes, and classes jointly, with a FocalNet-L (Yang et al., 2022) backbone. Fig. 2 shows the configuration actually used. The backbone emits four feature levels, **res2** at one quarter of the input resolution through **res5** at one thirty-second. For FocalNet-L, **res2** carries 192 channels and measures 120×68 pixels at the native input size, which makes it the finest level available to any head that needs to resolve a trunk a few pixels wide. A pixel decoder projects all four levels

to a common width of 256 and, in the two-stage setting used here, also produces the proposals that initialise the object queries, which the transformer decoder then refines with deep supervision. Layer counts, query counts, and the remaining values are listed in Appendix A.

We adopt this architecture unmodified except at the input stem, whose 7×7 convolution is widened from three to four channels to accept RGB-D input, and the classification head, which predicts 13 classes to match this study’s species set. Appendix A lists the complete configuration as instantiated, and Appendix B the backbone specification.

Backbone selection proceeded by ablation over ResNet-50, Swin-T, Swin-B, Swin-L, FocalNet-B, and FocalNet-L (Yang et al., 2022), together with a Mask R-CNN reference to establish that the query-based architecture is itself responsible for a large part of the performance. Because four-channel input cannot inherit three-channel pre-trained weights, RGB-D configurations are trained from scratch. From-scratch training here requires full-precision arithmetic, since mixed precision diverges within the first few hundred iterations. Appendix C records the settings this implies.

The Mask R-CNN reference is configured for the same regime rather than for fine-tuning, with a trainable stem and group normalisation, since frozen stages and frozen batch-normalisation statistics suit a pretrained backbone but leave part of a randomly initialised one at its initial weights. He et al. (2019) report the same requirement for detectors trained from random initialisation.

The ablations in Sections 3.2 and 3.3 and the final model use deliberately different protocols. Ablations run for 75 000 iterations with the diameter objective disabled, so that backbone and channel comparisons are made on segmentation alone at a budget that admits a full matrix of runs. The final model runs for 155 000 iterations with both objectives active. Each ablation is internally controlled, every arm within a table sharing schedule, data, optimiser, and seed, so comparisons inside a table are valid. Ablation values are nonetheless below what the same configuration reaches at the final model’s budget, an offset Section 3.3 quantifies.

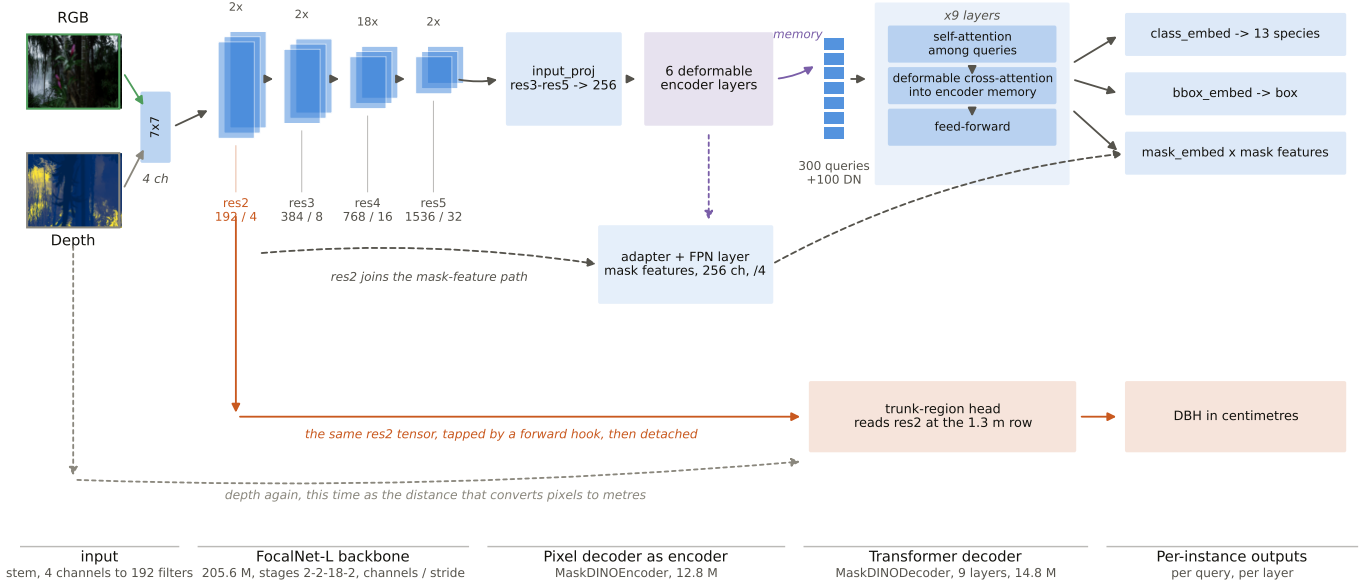


Figure 2: Architecture of the final model, drawn over a real SPREAD scene and its paired depth map. RGB and depth form one four-channel input to a single FocalNet-L backbone. Both heads read the same **res2** feature, the MaskDINO decoder unchanged and the trunk-region head with that feature detached, so the diameter loss never reaches the backbone (Section 2.2.4). Depth enters twice, as an input channel and as the geometry that converts pixel width into physical width.

2.2.3. Trunk-region diameter head

Diameter is predicted by an auxiliary head that reads a backbone feature directly rather than the decoder queries. The head taps **res2** through a forward hook on the backbone, so it consumes the same tensor the segmentation branch consumes rather than a recomputed copy, and the two objectives therefore share representation as well as parameters. The motivation is geometric. DBH is defined at a specific height on the trunk, so a head that pools features over an entire crown discards the localisation that the measurement requires.

Given an instance mask, the ground truth during training and the model’s own prediction at inference, and the depth map, we locate the image row whose back-projected world height is closest to 1.3 m,

$$h(r) = h_{\text{cam}} - \frac{(r - c_y) d(r)}{f_y}, \quad (2)$$

where r indexes image rows, $d(r)$ is the median depth along the masked part of that row, c_y and f_y are the principal point and focal length in pixels,

and h_{cam} is the camera height. Among the columns of the selected row we retain those closest to the camera, which isolates the trunk from foreground clutter sharing the same row, and take d_{trunk} as their mean depth and w_{px} as their count. This yields a purely geometric diameter estimate,

$$\hat{D}_{\text{geom}} = \frac{w_{\text{px}} d_{\text{trunk}}}{f_x}, \quad (3)$$

which we report as a baseline and also supply to the head as an explicit input feature. Fig. 3 traces these steps on one instance from the validation split.

Eq. 3 is the similar-triangles relation that Holcomb et al. (2023) and Feng et al. (2024) apply to the same measurement from mobile depth sensors. Both note that it treats the visible trunk edges as a chord rather than as tangents to a cylinder, and both propose a correction. At the pixel widths this imagery produces, the correction acts almost entirely on wide trunks, while most of the error comes from trunks narrower than 10 px, so we retain the uncorrected form.

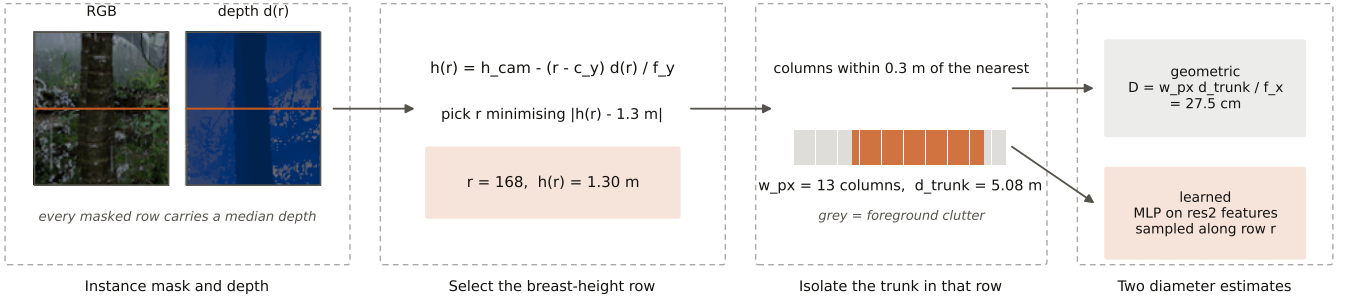


Figure 3: The steps of Section 2.2.3 traced on one validation instance, a BangaloPalm at 5.08 m. The selected row is 168, back-projected to 1.30 m, leaving 13 trunk columns and a geometric estimate of 27.5 cm. The learned head reads **res2** features along the same row, so the two estimates are not alternatives; the geometric value is itself an input to the head.

The head itself concatenates three sources: backbone **res2** features sampled from a strip of rows centred on the breast-height row, the geometric estimate of Eq. 3 together with d_{trunk} , and a species one-hot vector. A small multilayer perceptron, specified in Appendix D, regresses the diameter in per-species log space, which prevents the species with the largest stems from dominating the objective. Instances contribute to the diameter loss only if the geometry is well posed: at least five trunk columns, a back-projected height within 0.5 m of breast height, and a trunk distance inside the training range.

2.2.4. Multitask training

Both objectives share one backbone. A forward hook captures the **res2** feature map once per iteration, so the segmentation branch and the diameter head read identical features. The two gradients, however, are not treated alike. The captured feature map is detached before it enters the diameter head, so the head is fitted to the representation that the segmentation objective produces without contributing any gradient of its own to the backbone. Sharing is therefore one-directional. Section 4.1 reports the evidence behind this choice. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_{\text{DBH}} \mathcal{L}_{\text{DBH}}, \quad (4)$$

where $\mathcal{L}_{\text{seg}}$ is the standard MaskDINO combination of classification, mask, dice, box, and generalised IoU terms applied across all decoder layers and denoising queries, and $\mathcal{L}_{\text{DBH}}$ is a single scalar

regression term. This asymmetry matters for interpretation, since the segmentation objective is replicated over more than a hundred loss components while diameter contributes one, so nominally equal weights do not produce equal influence. We return to this in Section 3.4.

The final model is trained from scratch for 155 000 iterations at batch size four with AdamW, base learning rate 1×10^{-4} , weight decay 0.05, and step decay at iterations 124 000 and 144 667. The diameter term carries $\lambda_{\text{DBH}} = 5$, held constant for the whole schedule. Section 3.4 reports what happens when it is raised instead. Large-scale jittering at 640 px with scale range 0.4 to 2.0 is the only augmentation. The diameter loss admits instances out to 30 m, which covers 99.0 % of the depth distribution; the tighter 20 m bound that appears in the reported diameter numbers is a convention of the evaluation, not of training, and the two are kept distinct throughout.

2.2.5. Evaluation protocol

Segmentation is evaluated with COCO instance metrics. AP_{50} , the percentage of predicted masks whose overlap with a true tree (Intersection-over-Union, IoU) reaches at least 50%, averaged across classes and confidence thresholds, is the standard detection-accuracy score used throughout. Checkpoints are selected on the validation split and all reported results are measured on the held-out test split, except where a validation figure is named explicitly, which occurs only for the per-checkpoint training trajectories of

Section 4.1 and the geometric-baseline diagnostic of Section 4.3. Where a single operating point is required, we report precision, recall, and F1 at a stated score threshold rather than implying that a threshold-free number describes deployment.

Diameter is evaluated end-to-end. The head reads the model’s own predicted mask and predicted species, for instances the model detects and for which the predicted mask yields a valid breast-height geometry, rather than the ground truth. A predicted instance is matched to a ground-truth one at $\text{IoU} \geq 0.5$, greedy over prediction score, and both must then pass the geometric filters of Section 2.2.3 before the pair is scored. This is the condition a deployed system experiences, and it is what Section 3.1 reports as the main diameter accuracy. Where two configurations are compared directly, in Table 2 and Section 3.4, the comparison is restricted to instances for which both yield a diameter. A difference in which trees survive detection then cannot by itself produce a difference in the reported error.

We report root-mean-square error as the primary metric, because it appears in the large majority of the DBH literature and therefore permits comparison, with mean absolute error, bias, and coefficient of determination as supporting statistics. We additionally report the mean of per-species R^2 , since the aggregate value is inflated by one abundant species.

Distance-stratified analysis uses d_{trunk} from Eq. 2, which depends only on the ground-truth mask and the depth map and not on model weights. Identical instances therefore populate identical bins across models, and any difference between models within a bin is attributable to the models alone. Detection is matched class-agnostically at $\text{IoU} \geq 0.5$ so that a tree that is detected but misnamed is not counted as a miss, which separates two failure modes that a class-aware match would merge.

3. Results

We present the final model first, then the three ablations that justify its design, depth in Section 3.2, backbone in Section 3.3, and the diameter

loss weight in Section 3.4, before turning to where its errors lie.

3.1. The multitask model

The final model trains segmentation and diameter together from scratch for 155 000 iterations, with the diameter objective supervised on eleven species. Its architecture is the one shown in Fig. 2, one FocalNet-L backbone consuming four-channel RGB-D input, feeding the MaskDINO decoder and the trunk-region diameter head in a single pass, one network producing both outputs rather than two trained separately.

Evaluated end-to-end, using the masks and species the model predicts for its own detections, the final model returns a diameter for 27.1 % of eligible trees in the test split (1690 of 6233), with 7.62 cm RMSE, 4.89 cm MAE, a bias of -1.28 cm, and a mean per-species R^2 of 11.88 % (8 species with $N \geq 30$), at 62.29 % test AP_{50} . Section 2.2.5 defines the evaluation protocol in full, including how the comparisons of Table 2 and Section 3.4 control for which trees survive detection, and Section 4.5 discusses what the 27.1 % yield implies.

FocalNet-L was chosen for this role over two other backbones carried to the same multitask protocol. Table 2 compares all three end-to-end on the 1391 trees for which every backbone yields a diameter: FocalNet-B reaches a marginally higher AP_{50} (63.09 % against 62.29 %, within the run-to-run spread of Section 3.4), but FocalNet-L reaches a lower RMSE (7.40 cm against 7.74 cm) and a higher mean per-species R^2 (12.96 % against 9.03 %). FocalNet-L is used here because segmentation ties between the two while diameter does not. Section 3.3 describes how these two candidates were narrowed from six, and the further audit against Swin-L.

Fig. 4 places the annotation and the model’s own output side by side on one rainforest scene at a 0.30 score threshold, with 10 instances on each side so the comparison is qualitative, not a score. Panels (c) and (d) come from the same forward pass and the same set of queries, so a box and a mask on the same row of the output describe one instance. The boxes of neighbouring stems overlap heavily, which is what makes box-level

Table 2: FocalNet-L, FocalNet-B, and Swin-L at the final model’s own protocol, 155 000 iterations with the same stop-gradient and diameter objective. AP₅₀ and recall are test split, at the operating point of Table 4. Diameter metrics are end-to-end (Section 2.2.5) on the 1391 test trees all three backbones resolve on their own. Swin-L is an audit of a selection already made; see text.

Backbone	AP ₅₀ (%)	Recall (%)	RMSE (cm)	MAE (cm)	Bias (cm)	Mean per-species R^2 (%)
Swin-L	56.55	55.78	7.29	4.86	0.26	15.15
FocalNet-B	63.09	63.09	7.74	4.97	−1.64	9.03
FocalNet-L	62.29	61.72	7.40	4.79	−1.31	12.96

suppression unreliable under a closed canopy, while the masks of the same instances stay separable, the practical case for a query-based architecture in this setting. Two failure modes visible here are the ones Section 3.5 later quantifies.

Fig. 5 places the same four panels on a plantation scene with the same instance count, so the two are directly comparable. The contrast is visible without reading a number, the same contrast Section 4.2 quantifies (76.27 % AP₅₀ on plantation images against 45.98 % on rainforest).

Attaching the diameter objective is close to free on the segmentation side, which is a prerequisite for the multitask design to be worth adopting. Over the full schedule the two segmentation trajectories stay within run-to-run variation of each other (Fig. 6), so the second output is obtained without measurably degrading the first. What the coupling costs is not segmentation accuracy but the ability to optimise both objectives at the same checkpoint, quantified in Section 4.1.

3.2. Contribution of the depth channel

In the literature, comparisons of RGB against RGB-D are commonly confounded. A four-channel stem cannot inherit three-channel pretrained weights, so it is usually trained from scratch. The RGB arm, by contrast, starts from ImageNet initialisation. Any measured difference between the two arms then reflects the channel and the initialisation together, not the channel alone.

The direction and scale of this confound can be measured directly for FocalNet-L, the backbone used in the final model. Holding the channel count fixed, pretraining raises test AP₅₀ by 4.29 points on RGB (48.54 against 44.25) but costs 7.73 points on RGB-D once the three-channel stem is naively

inflated to four (49.45 against 57.18), since the ImageNet features that help a three-channel input work against a four-channel one. A comparison that pretrains both arms would therefore report an apparent depth advantage of only 0.91 points. Only when pretraining is removed from both arms does the 12.92-point advantage associated with depth become visible. Table 3 reports pairs trained from scratch under identical schedules, differing only in channel count.

Independent evidence that the confound is severe comes from work that does not control for it. Yu et al. (2026) segment tree trunks from real under-canopy RGB-D imagery and report an ablation in which architecture is held fixed and only the input changes. Both arms start from COCO-pretrained weights. Adding depth moves AP₅₀ from 83.9 to 84.0, a gain of 0.1 points, and mAP from 50.2 to 50.8. Read in isolation, that result suggests depth contributes almost nothing. Read against the measurement above, it is what a pretrained-on-both-arms comparison is expected to produce, since inflating a pretrained three-channel stem to four costs more than the depth channel returns. The two studies differ in task, data, and architecture and their numbers are not otherwise comparable, but the pattern is the one this section predicts.

Depth improves AP₅₀ in all six MaskDINO pairs, by 16.93 points for Swin-T, 15.93 for ResNet-50, 14.35 for FocalNet-B, 12.92 for FocalNet-L, 10.07 for Swin-L, and 9.68 for Swin-B. The Mask R-CNN pair moves by only 0.32 points, inside the run-to-run spread of Section 3.4, so that row is best read as showing no measurable effect in either direction. Both Mask R-CNN arms remain far below the MaskDINO family.

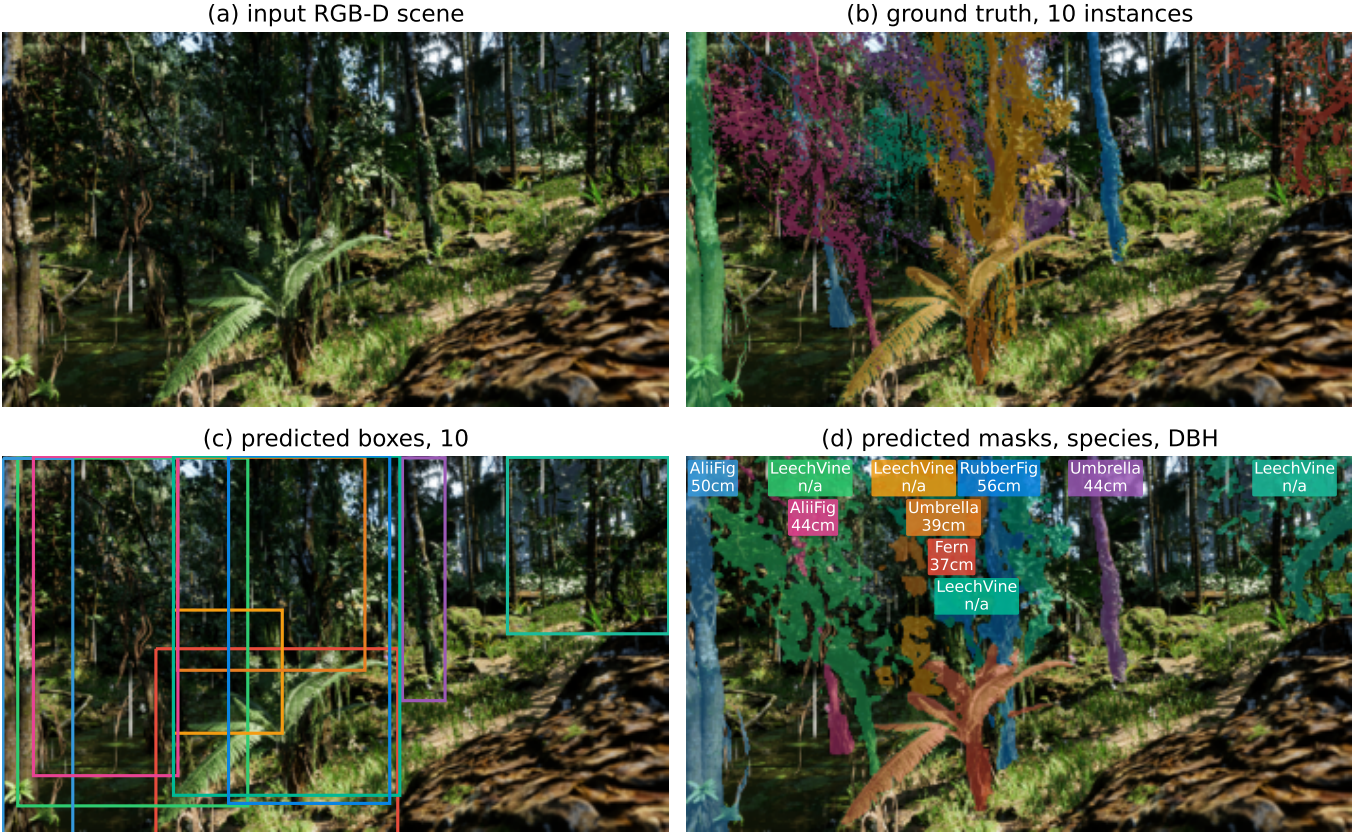


Figure 4: One rainforest scene from the test split, in four panels. (a) The input, an RGB image with its registered depth map. (b) The 10 annotated instances. (c) Predicted boxes and (d) predicted masks, both at score threshold 0.30 and from the same forward pass, with predicted species and diameter. Labels read *n/a* where the head found no valid breast-height row, typically an occluded trunk, or where diameter is not defined, as for LeechVine, a liana.

Table 3: Effect of the depth channel under matched initialisation. Both arms are trained from scratch for 75 000 iterations on identical data with the diameter objective disabled; only the input channel count differs. AP_{50} is the test-split value at the final checkpoint.

Architecture	Backbone	RGB	RGB-D	Δ
Mask R-CNN	ResNet-50	6.54	6.86	+0.32
MaskDINO	ResNet-50	32.00	47.94	+15.93
	Swin-T	33.91	50.84	+16.93
	Swin-B	38.95	48.63	+9.68
	Swin-L	40.75	50.82	+10.07
	FocalNet-B	43.75	58.10	+14.35
	FocalNet-L	44.25	57.18	+12.92

3.3. Backbone selection

Table 4 compares backbones under a common RGB-D from-scratch protocol at 75 000 iterations, to select among candidates rather than report converged accuracy. Under an identical recipe and

seed, FocalNet-L reaches 57.18 at 75 000 iterations and 62.29 at 155 000, so every entry lies roughly 5.11 points below its value at the final model’s budget. The table separates the FocalNet family from the rest by a margin far larger than the spread among the Swin variants or ResNet-50, but does not separate FocalNet-B from FocalNet-L, which differ by 0.92 points, inside the run-to-run spread of Section 3.4.

Both were therefore carried to the full multitask protocol used by the final model, 155 000 iterations with the same stop-gradient and diameter objective. Section 3.1 presents that comparison and the choice of FocalNet-L. Because that stage carried only the two backbones that segmentation could not separate, the selection procedure is itself budget-constrained. Swin-L, carried to the same protocol as an audit of it, matches on diameter (RMSE 7.29 cm) but its recall is 55.78 %, six to



Figure 5: One plantation scene from the test split, in the same four panels as Fig. 4 and carrying the same number of annotated instances, 10. Predicted boxes stay well separated instead of overlapping, and 9 of 11 detections receive a valid diameter, against several *n/a* in the rainforest scene. Nothing about the scene selection favours this outcome, since both were chosen only by matching instance count.

seven points below both FocalNet variants. Since detection rather than measurement precision limits how much of a stand the system reports on (Section 3.5), detecting fewer trees is not offset by measuring them slightly better. Swin-T scored within noise of Swin-L in the initial comparison (Table 4), so its omission from the full protocol is unlikely to change the outcome; Swin-B scored 2.19 points lower and is even less likely to. Neither was tested at the full schedule, an open limitation of the selection procedure. For segmentation alone, FocalNet-B matches FocalNet-L at roughly half the parameters, and the Mask R-CNN reference confirms that much of the performance belongs to the query-based architecture rather than the backbone.

3.4. Diameter loss weighting schedule

Because the segmentation objective is replicated across decoder layers while diameter contributes a single scalar, the diameter term supplies only 1.6 to 2.2% of the total loss at nominally comparable weights. One possible approach is to raise λ_{DBH} once segmentation has formed. We test it with a pair of runs sharing every setting except the weight schedule, the constant-weight run being the final model itself; the schedule holds $\lambda_{\text{DBH}} = 5$ to iteration 60 000, ramps linearly to 50 by 85 000, and holds thereafter. Their identical prefix provides a control, over which AP_{50} differs by between -2.90 and $+0.53$ percentage points, the evidence threshold for any difference measured later. Table 5 tracks segmentation loss at matched iterations with the diameter term excluded from both.

The gap grows monotonically, from 1.87% in

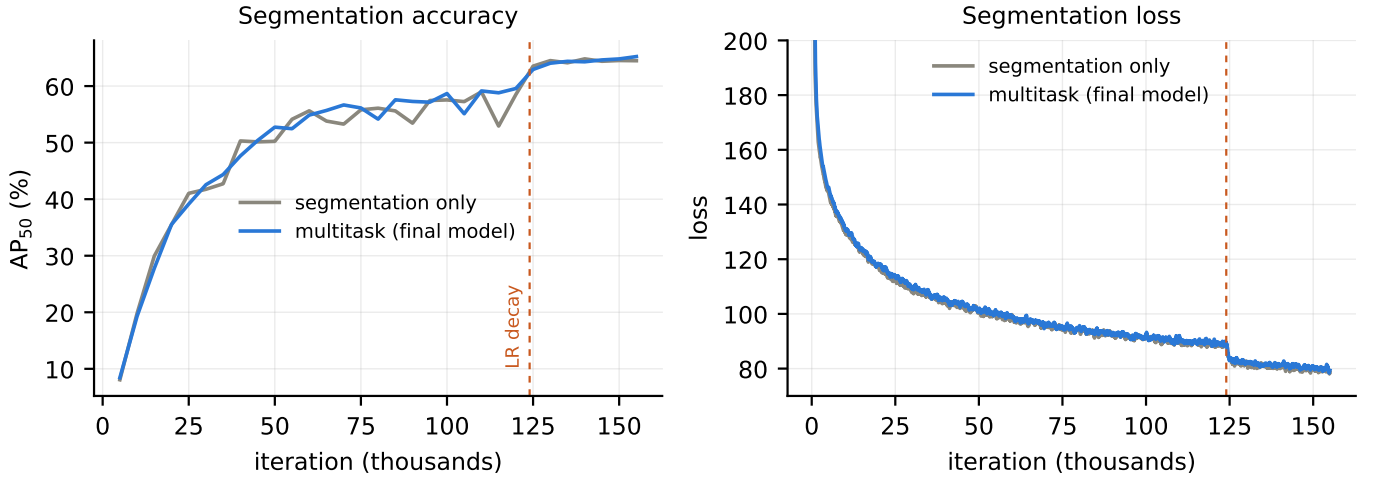


Figure 6: Training trajectories over 155 000 iterations, the multitask model against an otherwise identical segmentation-only run. The two curves remain within the run-to-run variation of Section 3.4 throughout. The step at iteration 124 000 is the first scheduled learning-rate decay. Curves are smoothed with a 25-point moving average and the diameter term is excluded from both, so the two measure the same quantity.

Table 4: Backbone comparison under RGB-D from-scratch training at 75 000 iterations, with the diameter objective disabled. AP_{50} is the test-split value at the final checkpoint. Recall is class-aware at score threshold 0.30 and $IoU \geq 0.5$. Params counts the whole network. *The two highest- AP_{50} backbones, carried forward to the full protocol (Table 2).

Architecture	Backbone	AP_{50} (%)	Recall (%)	Params (M)
Mask R-CNN	ResNet-50 (reference)	6.86	14.06	43.99
	ResNet-50	47.94	45.65	43.82
MaskDINO	Swin-T	50.84	48.73	52.96
	Swin-B	48.63	45.01	113.04
	Swin-L	50.82	46.52	222.80
	FocalNet-B*	58.10	58.02	113.91
	FocalNet-L*	57.18	57.02	233.17

the first pre-ramp window to 5.88 % once the schedule completes, against the roughly 2 % variation observed between the two runs before the schedule diverges. The schedule run plateaus near 99.3 from iteration 62 000 onward while the constant-weight run improves to 93.80, so segmentation does not deteriorate so much as stop improving. The unweighted diameter loss meanwhile shows no consistent benefit, being higher than or equal to the constant-weight run’s in five of six windows. A ten-fold increase in the diameter weight cost roughly 5.9 % in segmentation loss and returned no measurable improvement in diameter, which is consistent with the diameter gradient perturbing shared features rather than with a signal too weak to learn

from. The completed run confirms this end-to-end on the 1534 trees for which both runs yield a diameter, where the schedule run reaches 59.57 % test AP_{50} and 7.43 cm RMSE against 62.29 % and 7.37 cm for the constant-weight run. Raising λ_{DBH} costs segmentation without improving diameter.

3.5. Error stratification by camera distance

Aggregate scores describe a mixture. Because depth is an input, every ground-truth instance carries a physical distance computed from geometry alone, so the mixture can be decomposed without retraining and without letting the model influence the binning. Table 6 stratifies three abilities by

Table 5: Segmentation loss, excluding the diameter term, for the weighted-schedule run against the constant-weight run at matched iterations.

Iterations	λ_{DBH}	Schedule run	Constant run	Relative gap (%)
30 000–40 000	5	109.35	107.35	+1.87
45 000–55 000	5	103.68	101.58	+2.08
55 000–60 000	5	101.62	99.47	+2.16
62 000–68 000	≈ 14	100.11	97.50	+2.68
70 000–76 000	≈ 28	99.43	95.48	+4.13
80 000–86 000	41 to 50	99.32	93.80	+5.88

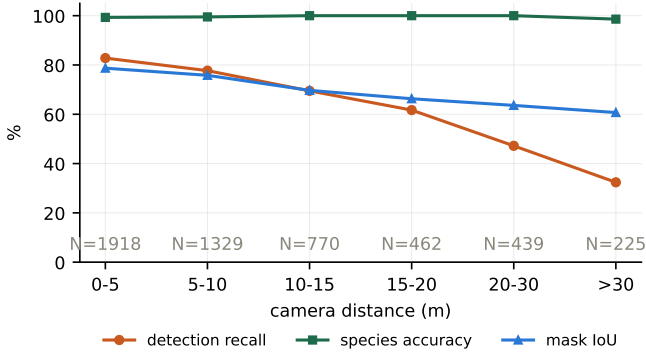


Figure 7: Three abilities stratified by camera distance, test split, computed from ground-truth masks and the depth map so that the binning does not depend on model weights. Detection recall falls sharply, mask IoU declines gradually, and species accuracy of the trees that are detected stays flat. Only detection limits the system.

that distance, and Fig. 7 plots the same three series so that their divergence is visible at a glance.

The three columns behave differently, and that difference is the finding. Recall falls by 58 % in relative terms from the nearest bin to the farthest. Mask quality falls by 23 %. Species accuracy falls by 0.7 % and is at or above 98.9 % in every bin, where in the last the residual is one instance out of 92 detections. If distance degraded the visual evidence to the point of ambiguity, species accuracy would fall with it; it does not. The trees that are detected remain identifiable and are still outlined reasonably well. What fails is detection.

Decomposing the failure confirms this. Of 7925 ground-truth instances, 4913 are detected and 4891 of those are also correctly named, so 3034 constitute the total failure. Of that failure, 3012 instances are simply not detected and 22 are de-

tected but misnamed. Misclassification accounts for 0.7 % of all failure, so the species classifier leaves essentially no room for improvement.

Two caveats apply. Species accuracy and mask IoU are conditional on detection, so the near-perfect classification in the far bins is measured only on distant trees large or unoccluded enough to be detected. The instances with no valid geometry are 33.8 % of the data, their recall of 42.4 % places them alongside the far bins, and the geometric filters that the diameter task applies remove exactly this hardest third from the diameter population.

3.5.1. Diameter error by distance

Applying the same stratification to diameter (Table 7) reverses the reading that absolute error invites.

Absolute RMSE rises from 6.96 cm to 10.39 cm, which reads as distance degrading the measurement. The mean ground-truth diameter rises over the same bins from 24.56 cm to 51.60 cm. Distant trees in this data are systematically larger, because small distant stems are the ones that fail to be annotated or to yield a valid geometry. Normalised by that, relative error is approximately flat between 20 and 28 %. The apparent distance dependence is largely a composition effect, and reporting absolute RMSE per bin without the accompanying mean diameter would invert the conclusion.

The learned head outperforms the geometric baseline of Eq. 3 in every bin, by roughly a factor of two, and retains that advantage at 20 m to 30 m, a range that the operating evaluation cap of 20 m excludes from all reported numbers.

Distance is a convenient axis but not itself

Table 6: Detection, classification, and mask quality by camera distance, test split, score threshold 0.30, matching at $\text{IoU} \geq 0.5$. Species accuracy and mask IoU are conditional on the instance having been detected. The final row groups instances whose breast-height geometry is ill-posed and which therefore have no assigned distance.

Distance	N	Recall (%)	Species correct (%)	Mask IoU (%)
0 m to 5 m	1889	82.4	99.6	79.1
5 m to 10 m	1375	79.3	99.7	75.8
10 m to 15 m	744	70.4	99.8	69.8
15 m to 20 m	472	59.5	100.0	65.4
20 m to 30 m	501	46.7	99.6	63.4
> 30 m	267	34.5	98.9	61.0
No valid geometry	2677	42.4	99.2	64.8
All	7925	62.0	99.6	72.2

Table 7: Diameter error by camera distance on the test split, with the mean ground-truth diameter of each bin. Bracketed intervals are bootstrap 95 % confidence intervals. The geometric baseline of Eq. 3 is shown for reference.

Distance	N	Mean GT (cm)	RMSE (cm)	Relative RMSE (%)	Geometric (cm)
0 m to 5 m	856	24.56	6.96 [6.22, 7.69]	28.3	15.22
5 m to 10 m	590	32.22	8.19 [7.36, 9.03]	25.4	15.44
10 m to 15 m	235	36.80	7.88 [6.46, 9.40]	21.4	17.60
15 m to 20 m	109	44.26	11.07 [8.71, 13.32]	25.0	20.33
20 m to 30 m	85	51.60	10.39 [8.01, 12.56]	20.1	20.36

the mechanism, since a distant trunk is harder mainly because more vegetation stands in front of it. Fig. 9 therefore re-bins the same detections by occlusion ratio, the share of an instance’s convex hull covered by instances nearer to the camera, traced on one validation instance in Fig. 8. Recall declines monotonically by 30.5 percentage points from barely occluded to heavily occluded stems, which places occlusion alongside distance as a direct driver of the detection shortfall rather than a consequence of it.

4. Discussion

Section 3.1 establishes that one network delivers both outputs without measurable cost to segmentation, and that it returns a diameter for only part of the stand. What that coupling costs, and what bounds diameter accuracy once it is paid, is the subject of this section.

4.1. Diameter accuracy under shared-feature supervision

Diameter accuracy in the multitask setting is bounded by something other than the amount of supervision the diameter head receives. Two pieces of evidence, both internal to the multitask runs themselves, indicate that the limiting factor is interference in the shared features.

First, the head is not short of optimisation; it stops improving long before training ends. In each of four multitask runs trained to 155 000 iterations, validation AP_{50} peaks at the final iteration while the best diameter RMSE arrives between 26 and 55 % of the schedule earlier, at iterations 114 999, 89 999, 74 999, and 69 999 respectively. The run without the stop-gradient of Section 2.2.4 shows the widest separation of the four. Its RMSE-optimal checkpoint (test RMSE 7.83 cm, bias -0.20 cm) occurs at 45 % of the schedule, while AP_{50} continues to rise from 51.06 % at that checkpoint to 64.05 % at the final one. This wider gap is not attributable to the distance cap this

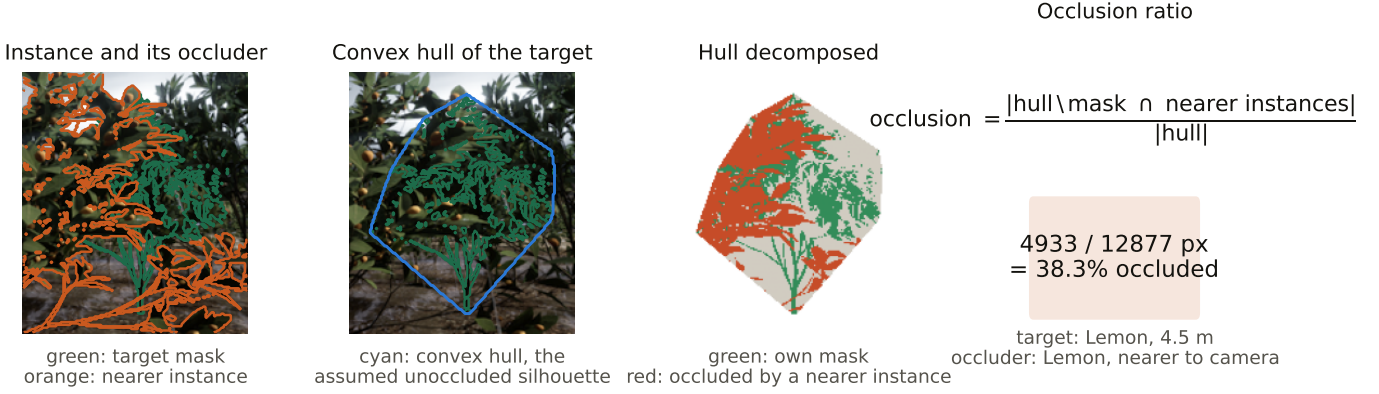


Figure 8: The occlusion ratio of Fig. 9 traced on one Lemon instance, 4.5 m from the camera and occluded by a nearer Lemon. The convex hull stands in for the unoccluded silhouette, and the share of it covered by nearer instances is the ratio, 38.3% here. The grey area belongs to neither mask and is this species’ sparse foliage.

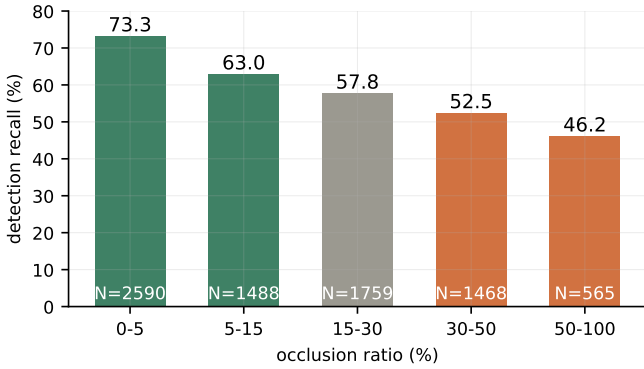


Figure 9: Detection recall against occlusion ratio, the share of an instance’s convex hull occupied by instances nearer to the camera. Recall declines monotonically by 30.5 percentage points from barely occluded to heavily occluded stems. The convex hull over-estimates the unoccluded silhouette for species with sparse, frond-like crowns, so the bins order occlusion severity rather than give the physical fraction hidden.

run uses for the diameter loss, since checkpoint-to-checkpoint AP_{50} noise between iterations 45 000 and 90 000 is comparable across all four runs (standard deviation 2.2 to 2.6 percentage points). After that point the diameter metric only oscillates. Across the 22 evaluation points beyond iteration 50 000 it moves within a band of 0.20 cm standard deviation once two outliers are excluded, while AP_{50} over the same span rises by 12.33 percentage points. The correlation between the two series is -0.04 . Further optimisation therefore continues to improve segmentation and does nothing for di-

ameter, which is not the behaviour of a head that is short of training.

Second, the two objectives cannot be served by one set of parameters at the same point in training. Because the optima are separated by tens of thousands of iterations, checkpoint selection becomes a trade-off rather than a joint optimum. In that run, moving from the last checkpoint to the diameter-optimal one improves RMSE by 0.35 cm at a cost of 12.99 percentage points of AP_{50} . A shared representation that served both tasks equally well would not produce separated optima.

A third line of evidence is indirect. Section 3.4 tests the response that follows most naturally from the hypothesis that the diameter signal is too weak, namely raising the diameter loss weight, and finds it counterproductive. A tenfold increase costs about 5% of segmentation loss and returns no improvement in the diameter loss. That outcome is what interference in the shared features predicts and a weak diameter signal does not.

Multitask training remains worthwhile. The diameter head currently shares the backbone’s step-decay schedule (Section 2.2.4) even though the backbone keeps changing under segmentation’s gradient, and whether a schedule decoupled from that decay lets the head track the shifting representation for longer is a question this study does not test.

4.2. Implications of the distance stratification

Species classification is highly accurate in this setting. It accounts for 0.7 % of total failure and does not degrade with distance. Mask quality declines gradually. Detection declines sharply and is the main limiting factor of the three.

This has a consequence for how such systems should be reported. A single recall figure of 62.0 % is a weighted mixture of 83 % nearby and 32 % far, with the weights set by the distance distribution of the evaluation data rather than by any property of the model. Two studies reporting the same aggregate number on stands of different density are not reporting the same thing. Distance-stratified reporting costs nothing beyond the binning and separates model quality from stand structure, so it is worth adopting wherever depth is available.

Real imagery corroborates that occlusion, not distance alone, drives the shortfall. Duclos et al. (2026) name tree occlusion, alongside species imbalance, as one of the two most pressing obstacles they encounter, which is the same conclusion Fig. 9 reaches here on rendered scenes.

The same pattern appears internally when the evaluation population is split by scene rather than by instance. Plantation images average 5.31 instances each against 12.47 for rainforest images, and segmentation on the final model tracks that difference closely. AP_{50} reaches 76.27 % on plantation images against 45.98 % on rainforest images, a gap of 30.29 points, with AR_{100} following the same direction (38.33 % against 27.74 %). This compares two scene types rather than isolating density in a controlled ablation, but its direction matches the instance-level occlusion result.

Whether that gap is intrinsic to rainforest scenes or a cost of training on both domains together can be separated directly. Under the protocol behind Table 3, specialising on plantation images alone reaches 74.06 % AP_{50} against 69.95 % for the jointly trained model on the same images, a cost of 4.11 points, while on rainforest images specialisation changes the result by 0.04 points. An independent ResNet-50 pair shows the same asymmetry. Within these experiments the rainforest deficit therefore appears to belong to the domain itself. Training both domains together gives up a

small, one-sided amount of accuracy rather than trading one domain for the other, which is why the final model is not split into domain specialists. A lightweight domain flag supplied at inference remains an untested middle ground.

The stratification also shows where the remaining improvement lies. Lowering the score threshold from 0.50 to 0.30 raises recall by 14.8 points in the nearest bin but by 27.9 points at 15 m to 20 m, which shows that many distant stems are detected with low confidence rather than missed outright. Part of the distance penalty is therefore a calibration effect and is recoverable by choosing an operating point deliberately, and part is genuine non-detection that a threshold cannot recover.

Finally, the third of instances whose breast-height geometry is ill-posed needs to be stated explicitly. Their recall places them alongside the most distant bins, so the diameter filters remove disproportionately hard cases. Diameter accuracy reported on the filtered population is valid for the population it describes, but it is not the accuracy a survey would experience across all stems in view. Section 4.5 carries this through to the model’s own predictions, where only 27.1 % of eligible trees receive a diameter at all.

4.3. Sensor resolution as a floor on diameter accuracy

Diameter accuracy in this study must be read against what the imagery can physically encode. With a focal length of 240 px at the native 480×270 resolution, the physical width subtended by a single pixel at distance d is d/f_x . Table 8 evaluates this.

At 20 m, the evaluation cap, a one-pixel difference in measured trunk width corresponds to 8.33 cm, which is the same magnitude as the end-to-end RMSE we report (7.62 cm). A method cannot express a trunk width more finely than the imagery quantises it, so a substantial part of the reported error is imposed before any model contributes. Because pixel width at a given distance scales as d/f_x , reaching a floor of 2 cm at this range, a plausible forestry-relevant target, needs $f_x \geq 1000$ px, roughly four times the current 240 px. Rendering at 1920×1080 rather than

Table 8: Physical width of one pixel at the native resolution, and therefore the granularity with which any method can express a trunk width at that distance. The last row is not measured but computed from the same relation, showing the pixel width that rendering at four times the native resolution would reach at the evaluation cap.

Distance	One pixel (cm)
5 m	2.08
10 m	4.17
20 m	8.33
30 m	12.50
20 m at 1920×1080 (4× native)	2.08

480×270 reaches 2.08 cm, and 2560×1440 leaves additional margin, at 1.56 cm. This gives a future SPREAD rendering pass a concrete target rather than a general call for higher resolution. The same relation explains the distance pattern of Table 7, since the pixel floor grows linearly with distance and so does the tolerable error, which is consistent with relative error staying flat while absolute error rises.

The prediction pattern in Fig. 10 is consistent with this floor. The least-squares slope of 0.74 shows regression towards the mean, small stems over-estimated and large stems under-estimated, which is the expected response of a regressor whose input cannot separate neighbouring diameters. Where the evidence is coarse, predicting near the conditional mean is the error-minimising choice, so the shrinkage reflects the quantisation the head works against rather than a defect of the head.

Resolution is not a limiting factor only in synthetic imagery. Duclos et al. (2026), working on photographed under-canopy scenes, likewise report substantial gains from higher image resolution and judge it likely to be fundamental to progress. Their measurement concerns detection and species identification rather than diameter, so the two findings are not the same claim, but they identify the same bottleneck from the opposite side of the synthetic-to-real divide.

Two consequences follow. Comparisons against studies reporting errors of a few centimetres are

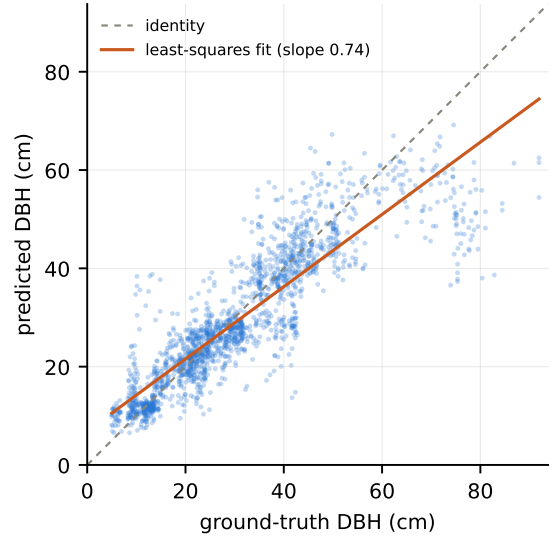


Figure 10: Predicted against ground-truth DBH for the final model. The least-squares slope of 0.74 shows regression towards the mean, small stems over-estimated and large stems under-estimated. This panel uses the full eleven-species population admitted out to 30 m ($N = 1875$), wider than the population behind the main error in Section 3.1, so its spread should not be read against that number directly.

not like-for-like unless their acquisition geometry is stated, since targeted close-range capture of one tree at a time operates far above this floor. And the most direct route to better diameter accuracy here is higher input resolution rather than a better regressor, which is a data generation decision rather than a modelling one.

4.4. Relation to prior work

Direct numerical comparison against published diameter errors requires care, because reported error depends on the sensing platform, the stand density, and the distance range admitted, and the studies reporting the smallest errors constrain acquisition most tightly. Close-range protocols reach a few centimetres because at one to two metres the pixel floor is roughly 0.5 cm, an order of magnitude finer than at this study’s 20 m cap, and Wu et al. (2025) photographs single rubber trees at that range with a physical reference marker. Terrestrial laser scanning (Liang et al., 2018) operates at a different level of accuracy altogether, while Grondin et al. (2022), the nearest structural precedent, likewise trains on synthetic renders and

reports error growing with camera distance. We therefore group comparisons by acquisition protocol rather than by a single reported number, and report our operating range explicitly.

Table 9 places published under-canopy segmentation results beside this study, with the reasons no pair is head-to-head shown next to the numbers. Within one study and protocol, Duclos et al. (2026) report AP_{50} of 90.8 without species and 51.2 with them, so requiring species costs 39.6 points on their own data. Against the one row sharing the species-aware task, this study is 11.1 points higher on AP_{50} but lower on the supporting metrics, by 12.4 points of AP and 18.4 of AR_{50} , matching what Section 3.5 measures internally, since at a loose overlap threshold the model is accurate on the trees it detects but does not detect enough of them. The evaluation populations differ enough that neither the advantage nor the deficits should be read as a ranking.

4.5. Limitations and future work

Synthetic training data. All training and evaluation use rendered imagery; no independent real-world evaluation is reported. No public under-canopy dataset shares SPREAD’s species composition, so any transfer test available today would confound domain shift, rendered against photographed imagery, with taxonomic shift, SPREAD’s thirteen species against an unrelated flora, leaving the result uninterpretable as evidence about either effect alone. Real-world performance therefore remains unvalidated, and closing this gap requires either a species-matched real dataset or fine-tuning data collected under the same taxonomic scope as SPREAD.

Detection under occlusion. Section 4.2 explains the mechanism behind the detection short-fall, but explaining a failure mode is not closing it, and the proportion of eligible trees that receive a diameter is far lower than the accuracy of the head alone suggests. Recall still falls to 34.5 % beyond 30 m, and rainforest segmentation, 45.98 % AP_{50} , remains far below plantation, 76.27 %. The 7.62 cm RMSE is the error on the 27.1 % of eligible trees that the model detects and for which it yields a breast-height row. Of 6233 eligible test trees,

66.9 % are matched to a detection at all; of those, only 40.5 % yield a usable diameter. A tree that is detected but whose mask cannot resolve its trunk is, for inventory purposes, indistinguishable from one never detected, and the two failure modes are comparable in scale.

Diameter definition and resolution. Diameter is not equally well posed across species. Four of the eleven supervised species are small fruit trees that branch at or below breast height, so a single diameter at 1.3 m is not the quantity a field crew would record for them, and each contributes fewer than 30 test instances. They are supervised anyway because excluding them would narrow what the model can report on. Separately, Section 4.3 shows that at the 20 m cap a single pixel already spans 8.33 cm, the same magnitude as the reported RMSE, which is also why predictions are systematically rather than randomly shrunk toward the mean (least-squares slope 0.74 in Fig. 10). Unlike the other limitations, this one is a data generation decision and is directly addressable in future rendering passes.

Future work. Three directions follow. First, collecting or annotating a real under-canopy dataset that shares SPREAD’s species composition would let a transfer test separate domain shift from taxonomic shift, and would allow fine-tuning on real imagery. Second, rendering SPREAD at higher resolution would raise the pixel floor directly, and Table 8 gives a concrete target for that pass. Third, an architectural change aimed at detection may be more direct than further refinement of species classification or diameter estimation, since detection accounts for 99.3 % of failure at long range while the other two already perform well on the trees that are detected. The finest feature level available to detection here, **res2** at one quarter of the input resolution, already discards spatial detail before a query forms. Amodal completion targets the occlusion mechanism Fig. 9 isolates, and aggregating evidence across nearby frames, rather than the single passive frame this study assumes, could recover a trunk that is hidden in one frame from another. None of these is tested here.

Table 9: Published segmentation results on under-canopy imagery, placed beside this study. **The rows are not head-to-head comparisons and no ranking is intended.** They differ in the four respects given in the left-hand columns, and the two class-agnostic rows show what the same metrics reach when species need not be distinguished. Dashes mark quantities the source does not report. Average recall is not defined identically across sources, so that column is indicative rather than exact.

	Imagery	Masks	Classes	AP ₅₀	AP	AR ₅₀	AR
This study	rendered	whole tree	13	62.29	26.79	70.56	33.45
Duclos et al. (2026), species-aware	photographed	trunk only	21	51.2	39.2	89.0	68.6
<i>Class-agnostic, shown for scale only</i>							
Duclos et al. (2026), trunk detection	photographed	trunk only	1	90.8	69.9	–	76.4
Mo et al. (2025)	photographed	trunk only	1	91.4	69.4	–	–

5. Conclusions

We set out to determine whether a single network can deliver the two measurements that forest inventory records, from the viewpoint a field crew occupies, and where such a system actually fails.

Depth is not a marginal input in this setting. Under controls that remove the usual confound with pretrained initialisation, adding a depth channel raises AP₅₀ by 12.92 percentage points for FocalNet-L, the backbone used in the final model, and up to 16.93 points across the six MaskDINO backbones tested. It is also what makes diameter recoverable at all, since converting an apparent trunk width to a physical one requires a distance.

A single multitask network reaches 62.29 % test AP₅₀ while producing both outputs in one pass, at no measurable cost to segmentation. Evaluated end-to-end on its own detections, it returns a diameter, 7.62 cm test RMSE, for 27.1 % of eligible trees; the shortfall is split roughly evenly between trees the model never detects and trees it detects but whose predicted mask is too imprecise to resolve a trunk. Diameter accuracy itself, however, is bounded by interference in the shared features rather than by the supervision the head receives, and raising the diameter loss weight, the intuitive response, degrades segmentation without improving diameter.

Stratifying by camera distance, which the depth input makes possible without retraining, shows that the three abilities the system needs behave very differently. Detection falls from 82.4 % to 34.5 % across the distance range and accounts

for 99.3 % of all failure. Species identification of detected trees holds at 98.9 to 100% throughout. Relative diameter error is approximately flat once the systematic increase in tree size with distance is accounted for. The system is limited by detecting stems, not by recognising or measuring them, so the classifier and the regressor leave little room for improvement.

CRedit authorship contribution statement

Akhmad Fauzi Nurulhamzah: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Muhamad Risqi Utama Saputra:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The SPREAD dataset used to train and evaluate the models in this study (Feng et al., 2025) is publicly available at <https://github.com/FrankFeng-23/SPREAD>. Code and trained model checkpoints are available from the corresponding author upon reasonable request.

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Appendix A. Segmentation architecture as instantiated

Section 2.2.2 states the architecture at the level a reader needs to follow the argument. Table A.10 records the complete configuration, so that the runs can be reproduced without access to our repository. Every value is read from the configuration file written by the final training run, not transcribed from a template. Where our value departs from the reference implementation of Li et al. (2023a) the difference is noted; unmarked rows are that implementation’s setting for instance segmentation.

Table A.10: MaskDINO architecture of the final model, as instantiated. *Ours* marks the rows where this work departs from the reference implementation: the input channel count is the architectural change, and the class count follows the dataset. Training settings continue in Table A.11.

Component	Setting	Value
<i>Input and backbone interface</i>		
Backbone	FocalNet-L, focalnet_L_384_22k	—
Input channels	FOCAL.in_chans	4 (<i>this study</i>)
Backbone outputs	res2-res5	192/384/768/1536
Strides	res2-res5	4/8/16/32
Pixel mean, fourth channel	PIXEL_MEAN[3]	118.1
<i>Pixel decoder</i>		
Module	MaskDINOEncoder	—
Deformable encoder layers	TRANSFORMER_ENC_LAYERS	6
Projection width	CONVS_DIM, MASK_DIM	256
Normalisation	SEM_SEG_HEAD.NORM	GN
<i>Transformer decoder</i>		
Module	MaskDINODecoder	—
Decoder layers	DEC_LAYERS	9
Object queries	NUM_OBJECT_QUERIES	300
Hidden width	HIDDEN_DIM	256
Attention heads	NHEADS	8
Feed-forward width	DIM_FEEDFORWARD	2048
Query selection	TWO_STAGE	true
Encoder-side prediction	INITIAL_PRED	true
Anchor initialisation	INITIALIZE_BOX_TYPE	bitmask
Deep supervision	DEEP_SUPERVISION	true
Classes	SEM_SEG_HEAD.NUM_CLASSES	13 (<i>this study</i>)
<i>Denoising</i>		
Mode	DN	seg
Denoising queries	DN_NUM	100
Noise scale	DN_NOISE_SCALE	0.4

The matching cost and the loss are configured with the same numerical values but are separate mechanisms: the cost decides which query is paired with which ground-truth instance, and the loss is then computed on that pairing. Matching is one-to-one and combines classification, box, and mask terms, following Li et al. (2023a); no non-maximum suppression is applied at any point, and inference reduces to a top- k selection over the query-by-class score matrix.

Table A.11: Training configuration of the final model, continuing Table A.10. The matching cost decides which query is paired with which ground-truth instance; the loss is then computed on that pairing. The two are configured separately even where their values coincide.

Component	Setting	Value
<i>Matching cost</i>		
Class	COST_CLASS_WEIGHT	4.0
Box, L1	COST_BOX_WEIGHT	5.0
Box, GIoU	COST_GIOU_WEIGHT	2.0
Mask, cross-entropy	COST_MASK_WEIGHT	5.0
Mask, dice	COST_DICE_WEIGHT	5.0
<i>Loss</i>		
Class	CLASS_WEIGHT	4.0
Box, L1 and GIoU	BOX_WEIGHT, GIOU_WEIGHT	5.0, 2.0
Mask, cross-entropy and dice	MASK_WEIGHT, DICE_WEIGHT	5.0, 5.0
No-object down-weighting	NO_OBJECT_WEIGHT	0.1
Points sampled per mask	TRAIN_NUM_POINTS	12 544
<i>Optimisation</i>		
Optimiser	ADAMW, weight decay 0.05	—
Base learning rate	BASE_LR	1×10^{-4}
Backbone multiplier	BACKBONE_MULTIPLIER	1.0
Warmup	WARMUP_ITERS	5000
Iterations	MAX_ITER, steps at	155 000; 124 000, 144 667
Batch size	IMS_PER_BATCH	4
Gradient clipping	full model, L2	0.01
Mixed precision	AMP.ENABLED	false (<i>this study</i>)
Stop-gradient, diameter to backbone	DBH_DETACH_BACKBONE	true (<i>this study</i>)
Distance bound, diameter loss	DBH_MAX_DEPTH	30 m (<i>this study</i>)
Diameter loss weight	DBH_JOINT_WEIGHT	5.0 (<i>this study</i>)

Appendix B. FocalNet-L backbone specification

FocalNet (Yang et al., 2022) replaces self-attention with focal modulation, which aggregates context around each position before interacting with the query rather than after. Because the mechanism is less widely known than self-attention, and because the diameter head reads `res2` directly, we record the instantiated backbone in full. Parameter shapes in Table B.12 are read from the trained checkpoint.

Table B.12: FocalNet-L as instantiated. Upper block: the four stages. Lower block: the parameter tensors of one focal modulation block, taken from the first block of stage one, where the width is 192.

Item	Detail	Value
<i>Stages</i>		
Embedding width	<code>embed_dim</code>	192
Blocks per stage	<code>depths</code>	2, 2, 18, 2
Focal levels per block	<code>focal_levels</code>	3
Focal window	<code>focal_windows</code>	5
Depth-wise kernels	$2k + 5$ for $k = 0, 1, 2$	5, 7, 9
Layer scale	<code>use_layerscale</code>	true
Post-modulation norm	<code>use_postln</code>	true
Modulator normalisation	<code>normalize_modulator</code>	false
<i>Patch embedding, stem</i>		
Convolution	kernel, stride, padding	7, 4, 2
Weight shape	<code>patch_embed.proj.weight</code>	(192, 4, 7, 7)
Parameters	weights and bias	37 632 + 192
Normalisation	<code>patch_embed.norm</code>	LayerNorm(192)
<i>One focal modulation block</i>		
Projection to q , context, gates	<code>modulation.f.weight</code>	(388, 192)
Focal level 1	<code>focal_layers.0.0.weight</code>	(192, 1, 5, 5)
Focal level 2	<code>focal_layers.1.0.weight</code>	(192, 1, 7, 7)
Focal level 3	<code>focal_layers.2.0.weight</code>	(192, 1, 9, 9)
Channel mixing of modulator	<code>modulation.h.weight</code>	(192, 192, 1, 1)
Output projection	<code>modulation.proj.weight</code>	(192, 192)
Feed-forward	<code>mlp.fc1</code> , <code>mlp.fc2</code>	(768, 192), (192, 768)
Layer scale vectors	<code>gamma_1</code> , <code>gamma_2</code>	(192) each

The projection width 388 is $2 \times 192 + (3 + 1)$: one copy of the input becomes the query, one becomes the context to be aggregated, and the remaining four channels are the gates that weight the three focal levels and the global average. The second dimension of every focal-level tensor is 1, which identifies them as depth-wise convolutions; the channel mixing that self-attention would perform is deferred to the 1×1 convolution that produces the modulator.

Because the three levels are applied in sequence rather than in parallel, the effective receptive field of the last level is wider than its kernel: $1 + (5 - 1) + (7 - 1) + (9 - 1) = 19$ feature cells, which at the stride of `res2` corresponds to roughly 76 pixels of the input image. This is the property that matters for under-canopy imagery, where a trunk is frequently occluded across part of its visible height and must be resolved from context that extends beyond the occluder, while the gating still allows the finest level to dominate at trunk boundaries.

Appendix C. Experimental environment

Table C.13 records the software and hardware used for every run reported in this paper, so that the results can be reproduced against the same versions. All training and evaluation ran on the MASSIVE M3 cluster under SLURM, one GPU per job.

Table C.13: Experimental environment and external libraries. Versions are those resolved in the conda environment used for the final model and for every ablation.

Category	Component	Version
Hardware	GPU	NVIDIA L40S (compute capability 8.9)
	Scheduler	SLURM, one GPU per job
	Cluster	MASSIVE M3
System	Operating system	Red Hat Enterprise Linux 9
	Python	3.9.23
Deep learning	PyTorch	2.1.0+cu121
	CUDA (PyTorch build)	12.1
	torchvision	0.16.0+cu121
	Detectron2	0.6
Supporting libraries	timm	1.0.26
	NumPy	1.26.4
	SciPy	1.13.1
	pycocotools	COCO evaluation and RLE masks

MaskDINO and FocalNet-L are used as released by their authors; the departures from those implementations are listed in Appendix A and shown as code in Appendix D. Correctness of each configuration change was checked by short smoke runs before the full schedule was submitted, rather than by a unit-test suite.

Appendix D. Original code

This appendix reproduces the two components that have no counterpart in the reference implementations, the trunk-region diameter head and the multitask training scheme that couples it to the backbone. Both were written for this project. Everything else in the training pipeline is MaskDINO and Detectron2 as released, driven by configuration files. Comments are omitted from the listings for brevity; line breaks and logic are otherwise unchanged.

Listing D.11 is the geometric core of the diameter head. It converts each masked row’s median depth into a world height using Eq. 2, selects the row closest to breast height, and then separates trunk columns from foreground vegetation by depth. It depends only on the mask, the depth map, and the camera intrinsics, which is what makes the distance stratification of Section 3.5 independent of model weights.

```
def _find_row_1p3m(mask, depth_raw, cy, fy, h_cam=_H_CAM, tol_wh=0.5, depth_tol=0.3):
    rows_with_mask = mask.sum(dim=1).nonzero(as_tuple=False).squeeze(1)
    if len(rows_with_mask) < 3:
        return None, None, None, None
    best_r, best_wh = None, None
    for r in rows_with_mask.tolist():
        cols = mask[r].nonzero(as_tuple=False).squeeze(1)
        if len(cols) < 2:
            continue
        d_vals = depth_raw[r][cols]
        valid = (d_vals > 0.1) & (d_vals < 200.0)
        if valid.sum() < 2:
            continue
        d = d_vals[valid].median().item()
        wh = h_cam - (r - cy) * d / fy
        if best_r is None or abs(wh - 1.3) < abs(best_wh - 1.3):
            best_r, best_wh = r, wh
    if best_r is None or abs(best_wh - 1.3) > tol_wh:
        return None, None, None, None
    row_cols = mask[best_r].nonzero(as_tuple=False).squeeze(1)
    row_deps = depth_raw[best_r][row_cols]
    valid = (row_deps > 0.1) & (row_deps < 200.0)
    row_cols, row_deps = row_cols[valid], row_deps[valid]
    if len(row_cols) < 2:
        return None, None, None, None
    d_min = row_deps.min().item()
    trunk_m = row_deps <= d_min + depth_tol
    trunk_cols = row_cols[trunk_m]
    d_trunk = row_deps[trunk_m].mean().item()
    if len(trunk_cols) < 2:
        return None, None, None, None
    return best_r, trunk_cols, d_trunk, best_wh
```

Listing D.11: Breast-height row selection and trunk-column isolation, from `scripts/trunk_roi_dbh_head_hybrid.py`.

Listing D.12 is the regressor itself. The three input sources described in Section 2.2.3, backbone features, the geometric estimate with its trunk distance, and a species one-hot vector, are concatenated into a single vector of width 207 for FocalNet-L. The final layer is initialised to zero weights and a bias of 3.5, which is the mean of the per-species log-diameter distribution, so that the head starts at the dataset mean rather than at an arbitrary value.

Listing D.13 is the multitask coupling. MaskDINO’s `forward()` is left untouched; a dynamically created subclass calls it, then adds the diameter term to the loss dictionary that Detectron2 already sums. A forward hook caches the `res2` feature map once per iteration, so both heads read the same features from a single forward pass. The stop-gradient of Section 2.2.4 is the single `detach()` call, which lets the head read those features without the diameter loss reaching the backbone.

```

class HybridTrunkROIDBHHead(nn.Module):
    def __init__(self, in_channels=192, hidden=256, strip_rows=1,
                  num_species=NUM_SPECIES, dropout=0.1):
        super().__init__()
        self.in_channels = in_channels
        self.strip_rows = strip_rows
        self.num_species = num_species
        feat_dim = in_channels + 2 + num_species  # backbone + geometric + species one-hot
        self.mlp = nn.Sequential(
            nn.LayerNorm(feat_dim),
            nn.Linear(feat_dim, hidden),
            nn.GELU(),
            nn.Dropout(dropout),
            nn.Linear(hidden, hidden // 2),
            nn.GELU(),
            nn.Dropout(dropout),
            nn.Linear(hidden // 2, 1),
        )
        nn.init.zeros_(self.mlp[-1].weight)
        nn.init.constant_(self.mlp[-1].bias, 3.5)

```

Listing D.12: The trunk-region diameter head, from `scripts/trunk_roi_dbh_head_hybrid.py`.

```

def _make_joint_class(base_cls):
    class _JointMaskDINOTrunkROI(base_cls):
        def forward(self, batched_inputs):
            losses = super().forward(batched_inputs)
            if not self.training:
                return losses
            device = next(self.parameters()).device
            res2 = self._res2_cache
            assert res2 is not None, "res2 hook not triggered"
            if self.dbh_detach_backbone:
                res2 = res2.detach()  # stop-gradient, Section 2.2.4
            depths, instances = [], []
            for x in batched_inputs:
                depths.append(x["depth"].to(device))
                instances.append(_prepare_dbh_targets(x["instances"].to(device)))
            dbh_loss, n_clean, n_total = self.dbh_head_trunkroi.compute_loss(
                res2, depths, instances, CAT_LOG_MEAN, CAT_LOG_STD,
                min_trunk_px=5, min_depth=1.0, max_depth=self.dbh_max_depth,
                max_wh_dev=0.5, species_weights=SPECIES_WEIGHT)
            losses["loss_dbh_trunkroi"] = dbh_loss * self._dbh_weight_now()
            return losses
    return _JointMaskDINOTrunkROI

```

Listing D.13: Multitask coupling and the stop-gradient, from `scripts/finetune_trunk_roi_dbh_hybrid_joint.py`. The weight returned by `_dbh_weight_now()` is constant in the final model and follows the schedule of Section 3.4 in the ablation.

Three further scripts produced the analyses rather than the model. `eval_end_to_end_dbh.py` implements the end-to-end diameter protocol of Section 2.2.5, `recall_per_jarak.py` produces the distance stratification of Table 6, and `occlusion_robustness.py` computes the occlusion ratio of Fig. 9. All three read stored predictions and ground truth only, so they require no GPU.

Appendix E. Dataset identification and partition

The dataset was not collected for this project. It is SPREAD (Feng et al., 2025), released publicly at <https://github.com/FrankFeng-23/SPREAD>. This study uses the ground-level under-canopy renders only, and retains the 13 species listed in Section 2.1.1.

The partition of Table 1 is made at the level of the rendered scene, not the individual image. Annotations are stored in COCO instance format with run-length encoded masks, extended with four custom per-instance fields beyond the COCO standard, `dbh`, `height_ue`, `crown_diameter`, and the per-instance depth statistics. These fields are not part of COCO and must be requested explicitly when the dataset is registered, otherwise the loader discards them silently and the diameter target never reaches the model. Depth maps are stored separately as single-channel PFM files in metres, paired with the colour images by filename.